THE PICARD FUNCTOR IS SMOOTH FOR RELATIVE DIMENSION AT MOST 1

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1790-1795 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1797-1810, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 22 of queue_moduli.json (pb=700, stmt 1790-1795, proof 1797-1810).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Transfer-type lemma: full logical chain displayed in-span; per-lemma credit notes name the engines.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DPJ: sources/stacks-moduli/pinned-original-source.tex, source lines 1583–1587. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DME: sources/stacks-moduli/pinned-original-source.tex, source lines 1639–1642. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AHD: sources/stacks-moduli/pinned-original-source.tex, source lines 1810–1810. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DN8: sources/stacks-moduli/pinned-original-source.tex, source lines 1810–1810. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Assume $f : X \to B$ has relative dimension ≤ 1 in addition to the other assumptions in this section. Then $\mathrm{Pic}_{X/B} \to B$ is smooth.*

Proof. By Lemma 0DPJ we know that $\mathcal{P}ic_{X/B} \to B$ is smooth. The morphism $\mathcal{P}ic_{X/B} \to \mathrm{Pic}_{X/B}$ is surjective and smooth by combining Lemma 0DME with Morphisms of Stacks, Lemma 0DN8. Thus if U is a scheme and $U \to \mathcal{P}ic_{X/B}$ is surjective and smooth, then $U \to \mathrm{Pic}_{X/B}$ is surjective and smooth and $U \to B$ is surjective and smooth (because these properties are preserved by composition). Thus $\mathrm{Pic}_{X/B} \to B$ is smooth for example by Descent on Spaces, Lemma 0AHD. □